

DS 4440 Embeddings

Today

- Our first foray into
representation learning
- How to encode TEXT
- Word2Vec (and beyond)

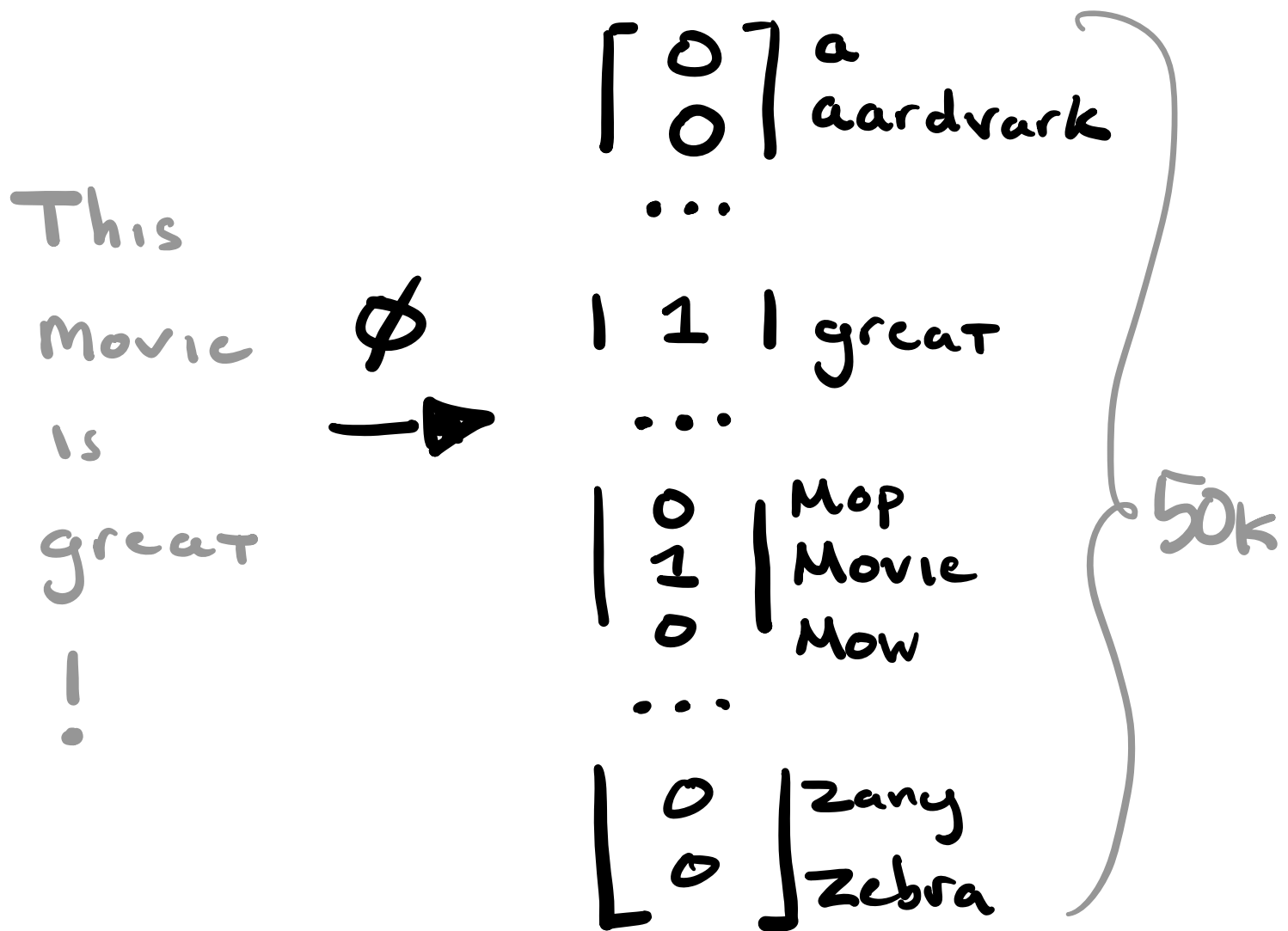
[  
"Cat"

[  
"Dog"

[  
"Pancake"

Wayyy back ~2013

Bag-of-Words (BoW)



Word w $\rightarrow$ Position w_{idx}

$$x_{w_{idx}} = \begin{cases} 1 & \text{if } w \text{ in input} \\ 0 & \text{otherwise} \end{cases}$$

Long, sparse $\times$ vectors.

Problems with BoW?

We often want to measure
Text Similarity

$$\text{Cos}(a, b) = \frac{\phi(a)^T \phi(b)}{\|\phi(a)\| \|\phi(b)\|}$$

For cat, dog

$$\begin{array}{c} \text{"cat"} \\ [0 \dots 1 \dots 0] \cdot \begin{array}{c} [0] \\ \dots \\ 1 \text{ "dog"} \\ \dots \\ 0 \end{array} = 0. \end{array}$$

So

$$\text{Sim}(\text{cat}, \text{dog}) =$$

$$\text{Sim}(\text{cat}, \text{pancake}) = 0.$$

Distributional Semantics

"You shall know a word
by the company it keeps"



Words to "low dimensional"
embeddings

Similar words ~ Nearby

Word2Vec [Mikolov 2013]

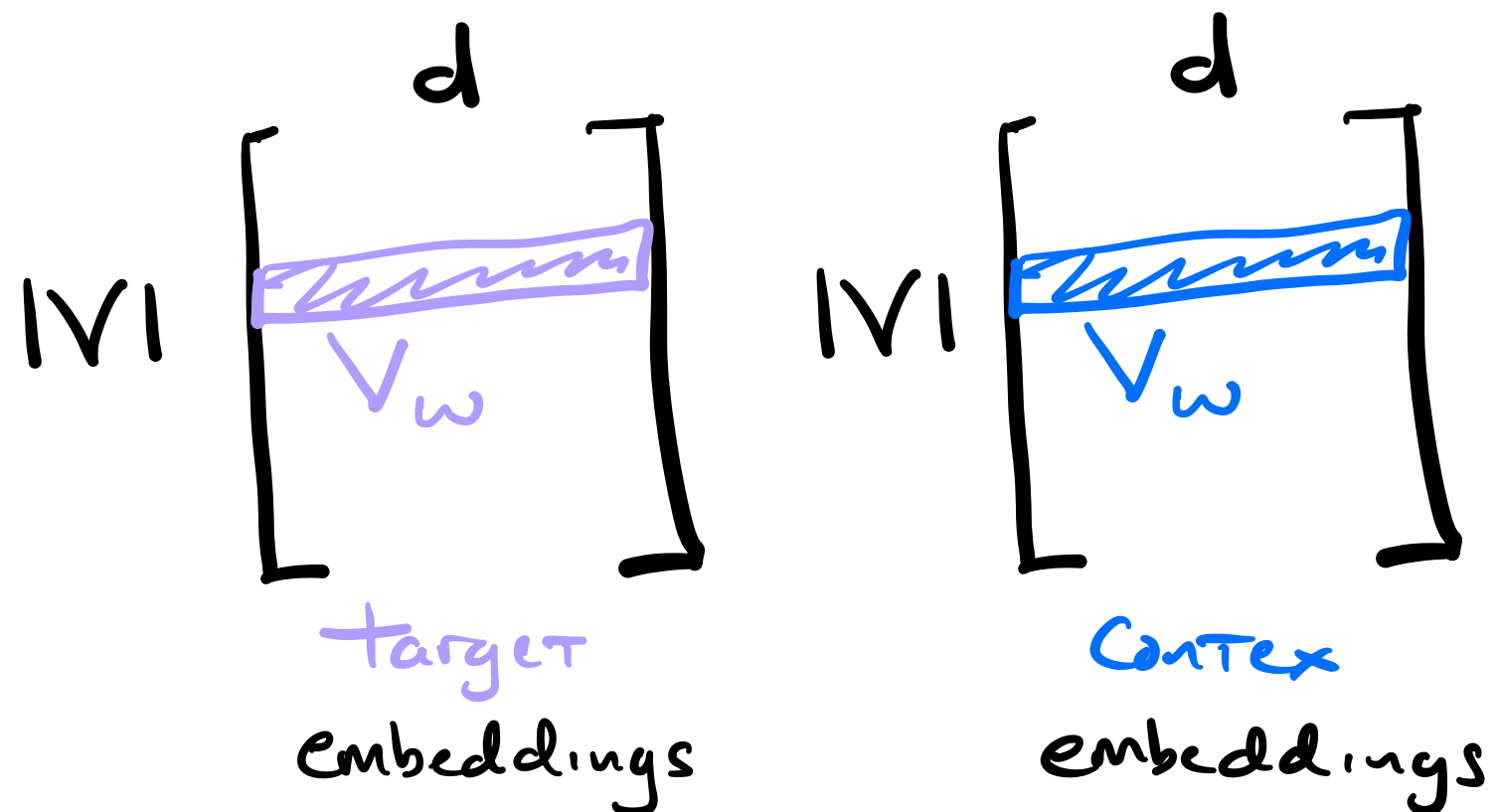
$$V_{\text{king}} - V_{\text{man}} + V_{\text{woman}} \approx V_{\text{queen}}$$

Target word and Context

Every word gets two vectors

V_w for when target (out)

u_w for when Input

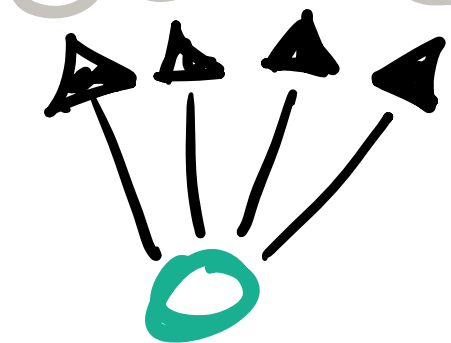


Two Variants / objectives

Skip-gram & CBOW

Skip-gram predicts Context words from target.

The man his son



loves

$P(\text{The} | \text{loves}) \cdot$

$P(\text{man} | \text{loves}) \cdot$

$P(\text{his} | \text{loves}) \cdot$

$P(\text{son} | \text{loves})$

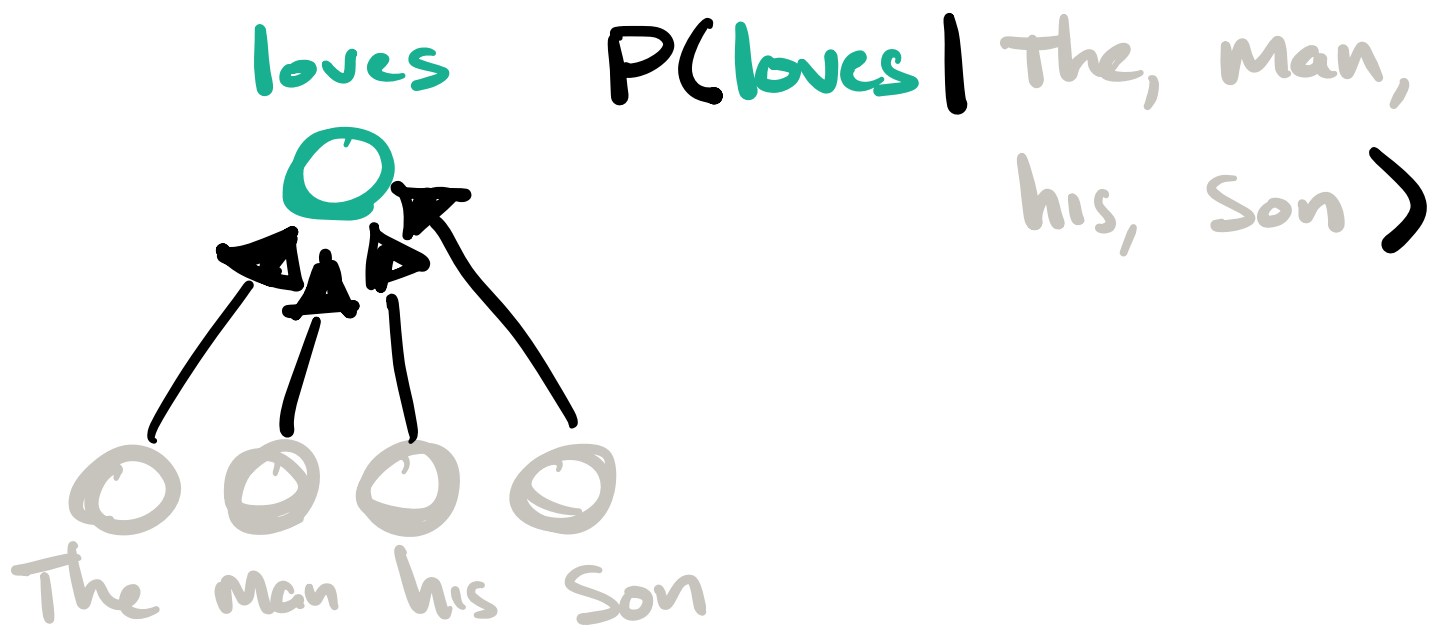
$$\text{Loss: } -\log \left\{ \prod_{w_t} \prod_{w_c \text{ in Context}} P(w_c | w_t) \right\}$$

$$= - \sum_{w_t} \sum_{w_c \text{ in Context}} \log P(w_c | w_t)$$

$$\Delta P(w_c | w_t) \equiv \frac{\exp\{\mu_{w_c} \cdot v_{w_t}\}}{\sum_{\tilde{w} \in V} \exp\{\mu_{\tilde{w}} \cdot v_{w_t}\}}$$

Softmax!

CBOW predict Target from Context words.



$$\mathcal{L}_{\text{loss}}: -\log(P(\omega_{\tau} | \omega_{c_1} \dots \omega_{c_k}))$$

$$\equiv -\log\left(\frac{\exp\{\mu_{\omega_{\tau}} \cdot \bar{v}_c\}}{\sum_{\tilde{\omega} \in V} \exp\{\mu_{\tilde{\omega}} \cdot \bar{v}_c\}}\right)$$

Where:

$$\bar{v}_c \equiv \frac{1}{k} \sum_{j=1}^k v_{\omega_{c_j}}$$

OK, BUT V is BIG!

→ Very Slow

Idea Negative Sampling

First Sample

$(\omega_{\tau}, \omega_{c_1} \dots \omega_{c_k})$

Real pair $\uparrow$ push up

$$\mathcal{L} = -\log \sigma(\mathbf{v}_{w_\tau} \cdot \mathbf{u}_{w_{cl}}) -$$

$$\sum_{j=1}^k \log \sigma(-\mathbf{v}_{w_\tau} \cdot \mathbf{u}_{\tilde{w}_j})$$

k random pairs; $\downarrow$ push down

Δ [Recall $\sigma(-x) = 1 - \sigma(x)$]

Let's see an example.

$$\mathcal{V} = \{\text{The, cat, sat, on, mat, pancake}\}$$

Sentence: "The cat SAT on The Mat"

(Target) Sampled Context

Negative words: on, pancake

$$\mathbf{v}_{\text{cat}} = [.5 \ - .2], \quad \mathbf{u}_{\text{sat}} = [.8 \ .4]$$

$$\mathbf{u}_{\text{pancake}} = [-.6 \ .7] \quad \mathbf{u}_{\text{on}} = [.1 \ .9]$$

Forward $\rightarrow$

$\mathcal{L} = -\log(.579)$

<u>Pair</u>	<u>y</u>	<u>v · u</u>	<u>$\hat{y} = \sigma(v \cdot u)$</u>	<u>$\mathcal{L}$</u>	<u>$\hat{y} - y$</u>
(Cat, Sat)	1	.32	.579	.546	-.421
(Cat, pancake)	0	-.44	.392	.497	.392
(Cat, on)	0	-.13	.468	.630	.468

$\mathcal{L} = -\log(1 - .468)$

From past lectures, we know

$$\frac{\partial \mathcal{L}}{\partial \hat{y}} = \hat{y} - y$$

$$(\text{Cat}, \text{Sat}) : .579 - 1 = -.421 = \frac{\partial \mathcal{L}}{\partial \hat{y}}$$

$$(\text{Cat}, \text{pancake}) : .392$$

$$(\text{Cat}, \text{on}) : .468$$

Update embeddings!

$$\frac{\partial \mathcal{L}}{\partial u_{\text{Sat}}} = (-.421) \cdot \left(\frac{\partial \hat{y}}{\partial u_{\text{Sat}}} \right)$$

$$= (-.421) \cdot \mathbf{V}_{\text{cat}} = (-.421) [.5 \ -2] \\ = [-.210 \ 0.084]$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{V}_{\text{cat}}} = [-.525 \ .527]$$

Let $\alpha = .1$. Then:

$$\mathbf{V}_{\text{cat}} [.5 \ -2] \rightarrow [.552 \ -.253]$$

$$\mathbf{u}_{\text{sat}} [.8 \ .4] \rightarrow [.821 \ .392]$$

$$\mathbf{V}_{\text{cat}} \cdot \mathbf{u}_{\text{sat}} \cdot 32 \rightarrow .355 \quad \checkmark \text{ More Sim}$$

$$\mathbf{u}_{\text{pancake}} [-.6 \ .7] \rightarrow [-.62 \ .708]$$

$$\mathbf{V}_{\text{cat}} \cdot \mathbf{u}_{\text{pancake}} \cdot 44 \rightarrow -.521 \quad \checkmark \text{ Less Sim}$$

(See Notebook!)

Ok, but what's an embedding layer, really?

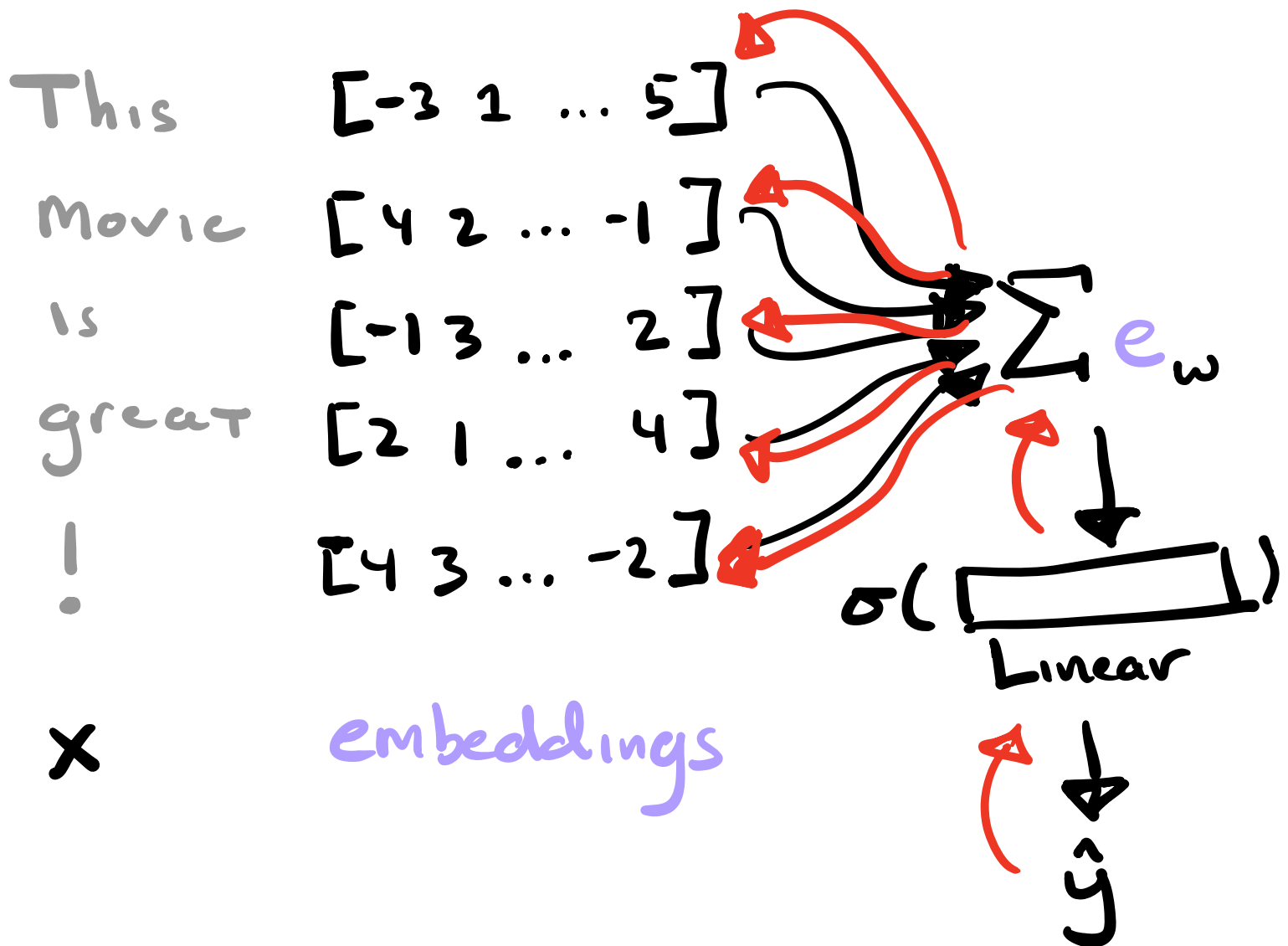
$$\mathbf{x}_w \in \mathbb{R}^{1 \times |V|}$$

$$\mathbf{E} \in \mathbb{R}^{|V| \times d}$$

$$\mathbf{e}_w = \mathbf{x}_w \mathbf{E} \quad \text{Selects row for } w.$$

How to use embeddings (for Text Classification)?

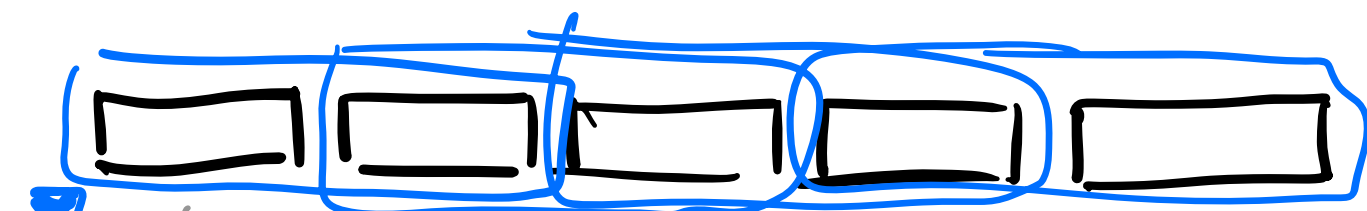
One way: Adopt CBOW



But! Averaging discards order!

So $\phi(\text{"not great"}) = \phi(\text{"great not"})$

A partial fix:



This movie is not great

"look" at two words
at a time.

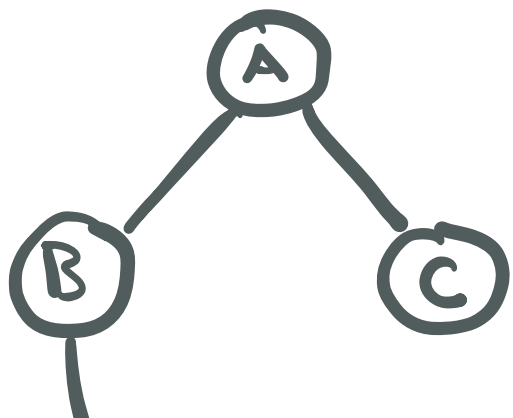
Convolution!

(HW & next lecture.)

Beyond "Words"

Embeddings not just for NLP!

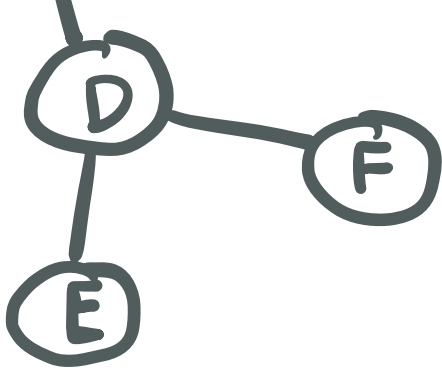
Example DeepWalk embeds
graph nodes (Perozzi et al.)



- Sample walks
randomly

- Treat as "c..."

Treat as sentences



A B D E

A C

...

B D F

Back to **NLP**, Can Treat
Paragraphs or documents as
Special "Words"

